

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA14

COURSE NAME: COMPILER DESIGN

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 25

Duration :60 Mins
On Class
Total Marks:50

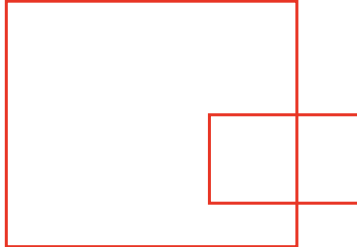
Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I sheik mohamed(192511145) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



Annexure A

SHORT ANSWER TEST

Q1. A lexical analyzer produces the following token stream:

id + id

Explain how the parser validates the syntactic structure of the expression.

Q2. Given the grammar:

$S \rightarrow aS \mid b$

Determine whether the string aaab belongs to the language using leftmost derivation.

Q3. Remove left recursion from the grammar:

$A \rightarrow Aa \mid b$

Q4. Find FIRST(S) for the grammar:

$S \rightarrow aA \mid b$

$A \rightarrow c \mid \epsilon$

Q5. Determine whether the grammar is suitable for top-down parsing.

$S \rightarrow Sa \mid b$

Justify your answer.

Q6. Apply left factoring to the grammar:

$S \rightarrow \text{if } E \text{ then } S$

$S \rightarrow \text{if } E \text{ then } S \text{ else } S$

Q7. Construct a parse tree for the string:

a + b

using the grammar:

$E \rightarrow E + E$

$E \rightarrow \text{id}$

Q8. Write the recursive descent procedures for the grammar:

$S \rightarrow aA$

$A \rightarrow b$

Q9. Given the grammar:

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$

Construct the predictive parsing table.

Q10. Demonstrate the first three actions of a shift-reduce parser for the input:

id + id

Q11. Identify the handle in the string:

id + id

for the grammar:

$E \rightarrow E + E$

$E \rightarrow \text{id}$

Q12. Construct the operator precedence relations for the operators:

+
*

where * has higher precedence than +.

Q13. Explain how operator precedence parsing handles the expression:

id * id

Q14. What is an LR parser? Illustrate the parsing process using the grammar:

$S \rightarrow aA$

$A \rightarrow b$

Q15. Given the grammar:

$S \rightarrow aA$

$A \rightarrow b$

Write the steps involved in constructing an SLR parsing table.

Q16. Compute FOLLOW(A) for the grammar:

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$

Q17. Construct the augmented grammar for:

$S \rightarrow a$

Q18. Find the closure of the LR(0) item:

$S' \rightarrow \bullet S$

for the grammar:

$S \rightarrow a$

Q19. Check whether the grammar is ambiguous:

$E \rightarrow E + E$

$E \rightarrow id$

using the string:

id + id + id

Q20. Construct a leftmost derivation for the string:

id + id

using the grammar:

$E \rightarrow E + E$

$E \rightarrow id$

Q21. Apply left factoring to the grammar:

$S \rightarrow aA$

$S \rightarrow aB$

$A \rightarrow b$

$B \rightarrow c$

Q22. Find FIRST(A) for the grammar:

$A \rightarrow x$

$A \rightarrow y$

Q23. Determine whether the grammar is suitable for predictive parsing.

$S \rightarrow aA \mid aB$

$A \rightarrow b$

$B \rightarrow c$

Justify your answer.

Q24. Construct recursive descent parser procedures for the grammar:

$S \rightarrow aS$

$S \rightarrow b$

Q25. Build the predictive parsing table for the grammar:

$S \rightarrow aS$

$S \rightarrow b$

Show the table entries for the terminals:

a, b, \$

1) $id + id$

Expression $\rightarrow$ Expression + Expression

The grammar is satisfy The expression is averted.

2)

$S \rightarrow as/b$

• $S \rightarrow as/b$

Left most derivation:

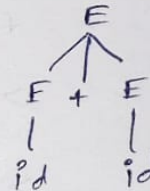
$S \rightarrow as$
 $\rightarrow aas$
 $\rightarrow aaas$
 $\rightarrow aaab$

6) Factored:

$S \rightarrow \text{if } E \text{ then } ss'$

$S' \rightarrow s/\epsilon$

7)



3)

$A \rightarrow Aalb$

result:

$A \rightarrow bA'$
 $A' \rightarrow aA'/\epsilon$

8)

procedure SC()

```

{
  match('a');
  A();
}
  
```

procedure A()

```

{
  match('b');
}
  
```

4)

FIRST(S)

$S \rightarrow aA/b$

$A \rightarrow c/\epsilon$

$FIRST(S) = \{a, b\}$

5)

$S \rightarrow salb$

NO, It contains left recursion, it not suitable for top-down parsing.

9)

Non-terminal

a

S

A

B

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$

16) $S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$

$FOLLOW(A) = b$

10) id + id

Shift id

Reduce id $\rightarrow E$

Shift +

17) $S' \rightarrow S$

$S \rightarrow a$

18) $S' \rightarrow \cdot S$

19) id + id + id multiple

parse tree

it is ambiguous

11)

It will handle the id

12)

operator

+

*

+

>

<

*

>

>

20) $E \rightarrow E + E$

id + E

id + id

13)

The first reduce id * id because * higher precedence operator.

21) $S \rightarrow aS'$

$S' \rightarrow A/B$

$B \rightarrow c$

$A \rightarrow b$

14)

LR parser:

LR parser bottom up parser, that scans left to right and produces a right most derivation.

22) $FIRST(A) = \{x, y\}$

23) NO, both reduction at begin causing parsing conflict.

15)

1) Grammar

2) LR(0)

3) LR(0)

4) First and Follow

5) Action

6) LR(0)

24)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand